



Transport substitution and travel resilience under bad weather: evidence from high-speed rail and air travel in China[☆]

Xinlei Hu^a , Shuping Wu^b, Longfei Zheng^{c,*}

^a School of Traffic and Transportation Engineering, Central South University, Changsha 410075, China

^b School of Economics and Management, Beijing Jiaotong University, Beijing 100044, China

^c School of Geography, Faculty of Geographical Science, Beijing Normal University, Beijing 100875, China

ARTICLE INFO

JEL code:

R41
L92
Q54

Keywords:

HSR-airsubstitution
Adverse weather
Intercity mobility
Climate resilience
Comprehensive transportation network

ABSTRACT

The growing frequency of extreme weather events presents challenges for transportation resilience. The substitution of different travel modes can enhance travel resilience under bad weather conditions. Using a novel dataset of nearly 3.7 billion high-speed rail (HSR) booking records in China from 2014 to 2016, this study estimates the extent to which HSR services are a substitute for air travel during bad weather conditions. Combining a difference identification strategy with high-dimensional fixed effects, we find that 2% of HSR trips (approximately 1.827 million trips) traveling over two hours can be attributed to air trips shifted to HSR during bad weather conditions. This weather-induced substitution is more pronounced for 2–7 h HSR trips and occurs mainly on weekdays and long holidays. The resilience of HSR to bad weather has generated welfare gains of approximately 151.61 to 400.13 million RMB for intercity travelers during 2014–2016.

1. Introduction

Facing the growing frequency of extreme weather events, high-speed rail (HSR) has emerged as a resilient alternative to the weather-sensitive aviation network for intercity travel. Extreme weather increasingly disrupts transportation systems worldwide (Zanni and Ryley, 2015). According to the World Meteorological Organization (WMO), the number of weather-related disasters has increased fivefold over the past five decades, with reported economic losses soaring from daily US\$ 49 million in the 1970 s to daily US \$ 383 million during 2010–2019 (WMO, 2021). Among major transport modes, aviation is particularly vulnerable due to its sensitivity to precipitation, wind, and reduced visibility. In 2022, weather-related factors accounted for 75% of air traffic disruptions in the United States (Climate Central, 2023), and extreme events such as Hurricane Helene in 2024 led to the cancellation of more than 1,000 flights.¹ In contrast, HSR systems exhibit substantially greater operational resilience and are less affected by adverse weather conditions (Bešinović, 2020; Chen et al., 2021; China Meteorological Administration (CMA), 2016; Zhou and Chen, 2020). Empirical studies have shown that under heavy rainfall conditions (25–50 mm/day), HSR systems in China maintain an on-time performance rate exceeding 95%, whereas flight cancellation rates exceed 40% under equivalent conditions (Federal Aviation Administration (FAA),

[☆] This article is part of a special issue entitled: 'Resilience & Regional Econ Dev' published in Transportation Research Part A.

* Corresponding author at: School of Geography, Faculty of Geographical Science, Beijing Normal University, No.19, Xijiekouwai Street, Haidian District, Beijing, China.

E-mail addresses: huxinlei1012@csu.edu.cn (X. Hu), shpwu@bjtu.edu.cn (S. Wu), longfeizheng@bnu.edu.cn (L. Zheng).

¹ For more details, please see <https://www.reuters.com/world/us/us-airlines-cancel-over-1000-flights-due-hurricane-helene-2024-09-26/>.

<https://doi.org/10.1016/j.tra.2026.105093>

Available online 4 June 2026

0965-8564/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1983; China Meteorological Administration (CMA), 2016; Li et al., 2024; Zhou et al., 2024). This resilience reflects key features of HSR infrastructure design, such as elevated tracks and distributed power systems, that mitigate flood and lightning risks (Janic, 2018; Bešinović, 2020), making HSR a credible substitute when air travel is disrupted by bad weather.

Previous studies have extensively explored the competitive dynamics between HSR and aviation (Albalade et al., 2015; Chen et al., 2017; Zhang et al., 2019). The introduction of HSR services tends to reduce passenger volume and airfares on parallel flight routes while improving service quality (Fang et al., 2024; Jiménez and Betancor, 2012; Ma et al., 2019). More importantly, these effects vary significantly across different intercity travel distances (Dobruszkes et al., 2014; Lawrence et al., 2019; Wan et al., 2016; Wang et al., 2018; Wu and Han, 2022). At the same time, an emerging body of literature emphasizes that HSR and air transport can also function as complementary modes within integrated intermodal systems, jointly shaping passenger travel behavior. Existing studies have examined factors influencing modal shifts, such as ticket pricing strategies, service frequency, passenger preferences, and station accessibility (Givoni and Dobruszkes, 2013; Wu and Han, 2022; Zhang et al., 2017). However, empirical evidence remains limited on how this interplay operates under external shocks such as adverse weather conditions, and how it contributes to the resilience of intercity travel systems.

This paper aims to address this gap by investigating HSR-air substitution during bad weather conditions in China. China provides an ideal setting for this analysis because its HSR system combines high operating speeds with extensive geographic coverage. By the end of 2024, China had developed over 48,000 km of HSR lines, with trains operating at speeds of up to 350 km/h and serving more than 80% of major cities. Specifically, we assemble a comprehensive dataset consisting of nearly 3.7 billion HSR ticket sales records, daily weather observations, and detailed geocoded information on HSR stations and airports across China from 2014 to 2016. We identify the causal effect using a difference strategy at the station-day level, designating HSR stations with airports in their service areas as the treatment group and other stations as the control group. By comparing passenger flow responses to bad weather across these groups, we quantify the number of travelers shifting from air to HSR. We further examine heterogeneity by travel distance, intra-city accessibility, and weather severity, and evaluate the calendar patterns and welfare implications of HSR-air substitution.

This paper makes three key contributions to the literature. First, this study is the first to provide national-level empirical evidence that coordination between HSR and air transport can significantly enhance intercity travel resilience under bad weather conditions. While theoretical and policy discussions have long emphasized the importance of developing integrated transport systems (Janic, 2001), empirical evidence on the realized benefits of intermodal substitution remains limited. Our empirical findings suggest that improving the accessibility between HSR stations and nearby airports can serve as an effective strategy for mitigating climate-induced travel disruptions.

Second, this study contributes novel evidence on how bad weather affects intercity travel behavior using high-frequency, full-sample travel data. Existing research has focused primarily on weather impacts on intra-city mobility (Huang et al., 2025; Wu et al., 2021), while intercity mode choices have remained largely unexplored (Lee et al., 2024; Sun et al., 2017; Xia et al., 2018). To address this, we leverage 3.7 billion full-sample HSR ticket sales records across China from 2014 to 2016 to identify the causal impact of bad weather on HSR-air substitution behavior. Unlike existing studies that rely primarily on theoretical models or localized case studies (Chen et al., 2021; Jiang et al., 2022; Li et al., 2022; Wang et al., 2018; Xia and Zhang, 2017; Xia et al., 2019; Yang and Zhang, 2012), our full sample data-driven analysis offers more representative and generalizable insights across diverse regions.

Third, this study advances the understanding of the broader benefits of HSR systems by uncovering their role in enhancing transport resilience. The literature has extensively documented that HSR promotes economic growth, facilitates intercity mobility, and reduces carbon emissions (Lin et al., 2021; Wang et al., 2019). By highlighting the operational stability of HSR under bad weather conditions, this paper reveals a previously underappreciated benefit whereby HSR enhances intercity travel resilience and passenger welfare. Our estimates suggest that the additional welfare gains brought by HSR under bad weather conditions amount to 151.61–400.13 million RMB during 2014–2016. Understanding these dynamics is essential for policymakers and operators to improve intercity travel resilience and passenger welfare, particularly in the face of increasingly frequent extreme weather events driven by climate change.

The rest of this paper is organized as follows: Section 2 describes the background and develops the research hypotheses. Section 3 introduces the data and empirical strategy, and Section 4 provides the baseline empirical results. Sections 5 and 6 summarize a set of heterogeneities and provide further discussions. Section 7 concludes.

2. Hypothesis development

Weather conditions play a crucial role in air transportation, as aviation operations are highly sensitive to environmental disruptions. Unlike ground transport systems, aircraft rely on precise meteorological conditions for safe takeoff, landing, and en-route navigation. Bad weather events, including heavy rainfall, strong winds, low visibility, and thunderstorms, can significantly impact airport operations and flight schedules (Chen et al., 2021; Gultepe, 2023). The literature has shown that moderate rainfall (daily cumulative precipitation of 10–25 mm) can reduce airport capacity by 20–30%, whereas heavy rainfall exceeding 25 mm per day often leads to extensive delays and cancellations (FAA, 1983; Li et al., 2024; Zhou, 2024). Crosswinds above 7 m/s can surpass aircraft operating limits, forcing runway changes or temporary closures (Civil Aviation Administration of China (CAAC), 2007). Additionally, low visibility below 800 m due to fog or snow can disrupt instrument landing system (ILS) operations, resulting in substantial flight delays or diversions (CAAC, 2001). These weather-related disruptions are not confined to individual airports but often propagate across the entire air traffic network, amplifying delays at both the regional and national levels (Wandelt et al., 2025). With climate change leading to an increased frequency and intensity of extreme weather conditions, the vulnerability of the aviation sector is expected to increase.

HSR has emerged as a reliable substitute for air travel under bad weather conditions. The global development of HSR began with Japan's Shinkansen in 1964, and was followed by significant network expansions in Europe, particularly in countries such as France, Germany, and Spain (Yoo et al., 2024). In comparison, China entered the HSR sector later but has experienced the fastest growth globally (Long et al., 2018). This pattern makes China an ideal case for examining HSR-air substitution. The country's first dedicated HSR line with operational speed of 350 km/h was introduced in 2008 (Zheng et al., 2025). By the end of 2024, China has built more than 48,000 km of HSR lines,² accounting for over two-thirds of the global network (Zheng et al., 2024). China's HSR system combines high operating speeds (250–350 km/h), extensive geographic coverage, and frequent service, making it highly competitive with aviation for medium- and long-distance travel (Lawrence et al., 2019; Wu and Han, 2022). The continuous expansion of the HSR network has led to a progressive shift in China's intercity travel markets, with HSR capturing a growing share of passengers who would have otherwise traveled by air (Chang et al., 2022).

Beyond competition and substitution, a growing body of literature emphasizes the cooperative relationship between HSR and air transport within integrated intermodal systems. Rather than functioning as purely competing modes, HSR and aviation can act as complementary services by reallocating demand across networks, improving system-wide efficiency, and enhancing travel resilience (Janic, 2001; Givoni and Banister, 2006; Dobruszkes et al., 2014). One prominent strand of this literature examines air-rail intermodality, where HSR is integrated with airport operations to replace short-haul feeder flights, reduce congestion at major hubs, and expand airport catchment areas (Clewlow et al., 2012; Givoni and Dobruszkes, 2013; Bešinović, 2020). Empirical evidence from Europe shows that coordinated air-rail services can improve connectivity, lower operating costs for airlines, and reduce carbon emissions without compromising passenger accessibility (Albalade et al., 2015; Behrens and Pels, 2012).

Compared with aviation, HSR is significantly less affected by bad weather conditions, making it a more resilient mode of intercity transportation (Chen et al., 2021). More recent studies highlight that HSR-air cooperation also plays an important role in enhancing transport system resilience. By providing a reliable ground-based alternative during disruptions such as extreme weather, strikes, or airport capacity constraints, HSR can absorb displaced air passengers and stabilize intercity mobility (Bešinović, 2020; Chen et al., 2021). Modern HSR infrastructures are designed to withstand heavy rainfall, strong winds, and low visibility with minimal operational impact, as shown in Table 1. Chinese railway regulations permit HSR operations at full speed under rainfall up to 50 mm per hour, conditions under which many flights would already face severe delays or cancellations. Even under extreme weather conditions, HSR trains can continue operating at reduced speeds.

As climate change increases the frequency of extreme weather events (WMO, 2021), the resilience of HSR offers a critical advantage in ensuring reliable intercity mobility. By providing a more stable travel option under bad weather conditions, HSR strengthens its competitive edge over aviation. This observation leads to our first hypothesis.

Hypothesis 1. During bad weather conditions, the number of passengers at HSR stations with airports within their service areas increases due to flight disruptions and delays.

HSR-air competition and substitution vary significantly with travel distance, a relationship well documented in the transportation literature (Dobruszkes et al., 2014; Wan et al., 2016; Wang et al., 2018). HSR has a competitive advantage over aviation for short- to medium-distance routes, whereas air travel remains dominant for long-distance routes (Lawrence et al., 2019; Wu and Han, 2022; Zhang et al., 2019). Specifically, for routes under 500 km (or 2 h by HSR), HSR dominates the market, capturing most passenger demand (Lawrence et al., 2019). For distances between 500 and 1,000 km (HSR 2–4 h), HSR holds a strong competitive edge over aviation because of its higher frequency, shorter boarding times, and direct city-center access (Chen, 2017). The most intensive competition between HSR and aviation occurs within the 1,000–1,750 km range (HSR 4–7 h), as the speed advantage of air travel begins to offset the disadvantages of longer boarding times and lower intra-city accessibility (Ma et al., 2019). This competition is exemplified by China's Beijing-Shanghai corridor (1,318 km), where both air and HSR services operate at high frequencies. The flight duration between the two cities is 2.5 h, while the fastest HSR service takes 4.5 h. However, when factoring in airport processing times and intra-city travel, the total door-to-door travel times of the two modes are comparable. Beyond 1,750 km (7 h by HSR), aviation generally maintains a significant advantage, as its substantially shorter flight times outweigh the longer airport procedures. For example, on long routes like Beijing-Guangzhou (2,298 km), air travel accounts for 84% of the market share,³ as the 8-hour HSR journey fails to compete with the time savings offered by aviation. In addition to travel time considerations, other factors such as comfort and schedule flexibility may also influence passengers' mode choices, particularly for long-distance trips. For example, HSR services typically offer more spacious seating and smoother travel experiences, which may be preferred by some travelers despite longer journey times. Moreover, the availability of overnight or sleeper HSR services can partially offset time disadvantages by allowing travelers to travel during nighttime, thereby reducing the opportunity cost of travel time. Given these variations in competition and substitution across travel distances, we propose that:

Hypothesis 2. During bad weather conditions, changes in passenger volume at HSR stations vary with travel distance.

The impact of bad weather on flight delays also varies significantly depending on the severity of weather conditions, as documented in Table 1. While minor weather disturbances may cause only slight schedule disruptions, more extreme conditions often lead to substantial delays or cancellations. The literature has shown that under moderate rainfall (10–25 mm per day), the probability of flight delays increases by 20–30%, with an average delay time of 30–75 min in China. In contrast, heavy rainfall exceeding 25 mm per day results in 50–70% of flights experiencing delays, with an average delay time of 50–120 min. When rainfall exceeds 50 mm per day,

² Data are from the Ministry of Transport of China (https://xxgk.mot.gov.cn/2020/jigou/zhghs/202506/t20250610_4170228.html).

³ Data are from the Civil Aviation Data Analysis Data Hub (<https://www.cadas.com.cn/>).

Table 1
Operational resilience of HSR and aviation during bad weather.

Weather conditions	Aviation impact	HSR impact	Data source
Panel A: Rainfall intensity			
10–25 mm/day (Moderate)	Delay probability: 20–30%; Average delay: 30–75 min	Full-speed operation	CMA, 2016; FAA, 1983
25–50 mm/day (Heavy)	Cancellations: 40–60%; Average delay: 50–120 min	Speed reduction: 200–300 km/h	
>50 mm/day (Torrential)	Cancellations: 60–90%	Speed reduction: 160–200 km/h	
Panel B: Crosswind speed			
5.5–12.9 m/s	Delays: 20–40 min	Full-speed operation	CAAC, 2007; CMA, 2016
12.9–20 m/s	Cancellations: >80%	Speed reduction: 200–300 km/h	
Panel C: Visibility			
800–1,600 m	Twin-engine aircraft grounded; Multiengine flights delayed	Full-speed operation	CAAC, 2001; CMA, 2016
200–800 m	Tri-engine and larger aircraft grounded	Full-speed operation	

Notes: For visibility < 800 m, aviation operations are restricted to aircraft equipped with ILS Category III systems, which enable automated landings with a runway visual range (RVR) as low as 50 m (ICAO Annex 10). Only certified aircraft and crews may operate under such conditions. CMA, FAA, and CAAC refer to the China Meteorological Administration, the Federal Aviation Administration, and the Civil Aviation Administration of China, respectively.

flight cancellations become significantly more frequent, and delays can extend beyond 120 min (FAA, 1983; Li et al., 2024; Zhou, 2024). Wind speed and visibility further exacerbate these disruptions, particularly crosswinds, which have a substantial impact on aircraft takeoff and landing (CAAC, 2001; Civil Aviation Administration of China (CAAC), 2007). Unlike aviation, which faces strict operational limits under bad conditions, HSR continues to provide a relatively stable alternative (China Meteorological Administration (CMA), 2016). As the weather severity increases, the likelihood of passengers switching from air travel to HSR travel increases accordingly. This leads to the last hypothesis:

Hypothesis 3. As weather conditions deteriorate, the number of passengers switching from air travel to HSR travel increases.

In addition to inter-city travel distance and weather severity, intra-city accessibility between airports and HSR stations plays a critical role in transportation substitution during bad weather. When the accessibility between an airport and the nearest HSR station is high, the cost and inconvenience of switching from air to HSR travel are substantially reduced. This improved accessibility makes HSR a more feasible substitute for passengers seeking to avoid long delays or cancellations. Conversely, poor accessibility may discourage such switching, even when flight disruptions are severe. Therefore, we propose the following hypothesis:

Hypothesis 4. The greater the intra-city accessibility between HSR stations and airports is, the greater the volume of passengers switching from air travel to HSR during bad weather conditions.

3. Data and empirical strategy

3.1. Data and measurement

3.1.1. Ticket sales records data and daily HSR passenger flow

The HSR passenger flow data used in this paper are calculated from actual ticket sales records, capturing all HSR trips in China from 2014 to 2016, for a total of nearly 3.7 billion travel records. As of the end of 2016, a total of 774 railway stations in China sold tickets for HSR services, identified by train numbers starting with G, D, or C. Each original ticket sales record includes information on the departure and destination stations, departure date, and travel time. By aggregating individual trip-level records, we construct the daily station-level passenger departure volume and further decompose them based on travel time.

We focus on HSR trips with a duration of more than two hours, where HSR-air substitution is most likely to occur under bad weather conditions. To capture variation in the intensity of HSR-air competitive substitution across different travel distances, we classify HSR trips into four groups based on travel time: 0–2 h, 2–4 h, 4–7 h, and more than 7 h (Lawrence et al., 2019; Wu and Han, 2022; Zhang et al., 2019). For HSR trips lasting 0–2 h, the corresponding travel distance is within 500 km, where HSR dominates with little to no competitive substitutions from air travel (Lawrence et al., 2019). Therefore, we exclude these short trips from our empirical analysis, as they are unlikely to capture HSR-air substitution behavior during weather disruptions.

Within the remaining sample of trips exceeding two hours, we further divide observations into three duration-based groups to test Hypothesis 2 and explore heterogeneity in substitution effects. Specifically, for trips lasting 2–4 h, HSR travel holds a competitive edge over air travel (Chen, 2017). Trips with a duration of 4–7 h represent the most intensive HSR-air competition. As travel distance increases, the competitive advantage of HSR over air travel diminishes (Ma et al., 2019). For trips exceeding 7 h, air travel takes the lead in the competition with HSR travel. This classification enables us to examine how substitution effects vary by travel distance during bad weather conditions.

Fig. 1 presents the spatial distribution of HSR stations and their passenger flows (for trips longer than 2 h) in 2016. The 2016 ticketing dataset includes a total of 1.44 billion HSR trips. Among them, 0.49 billion trips lasting more than 2 h are included in our analysis, as shown in Fig. 2. Passengers traveling for more than 2 h are primarily in politically significant and economically developed

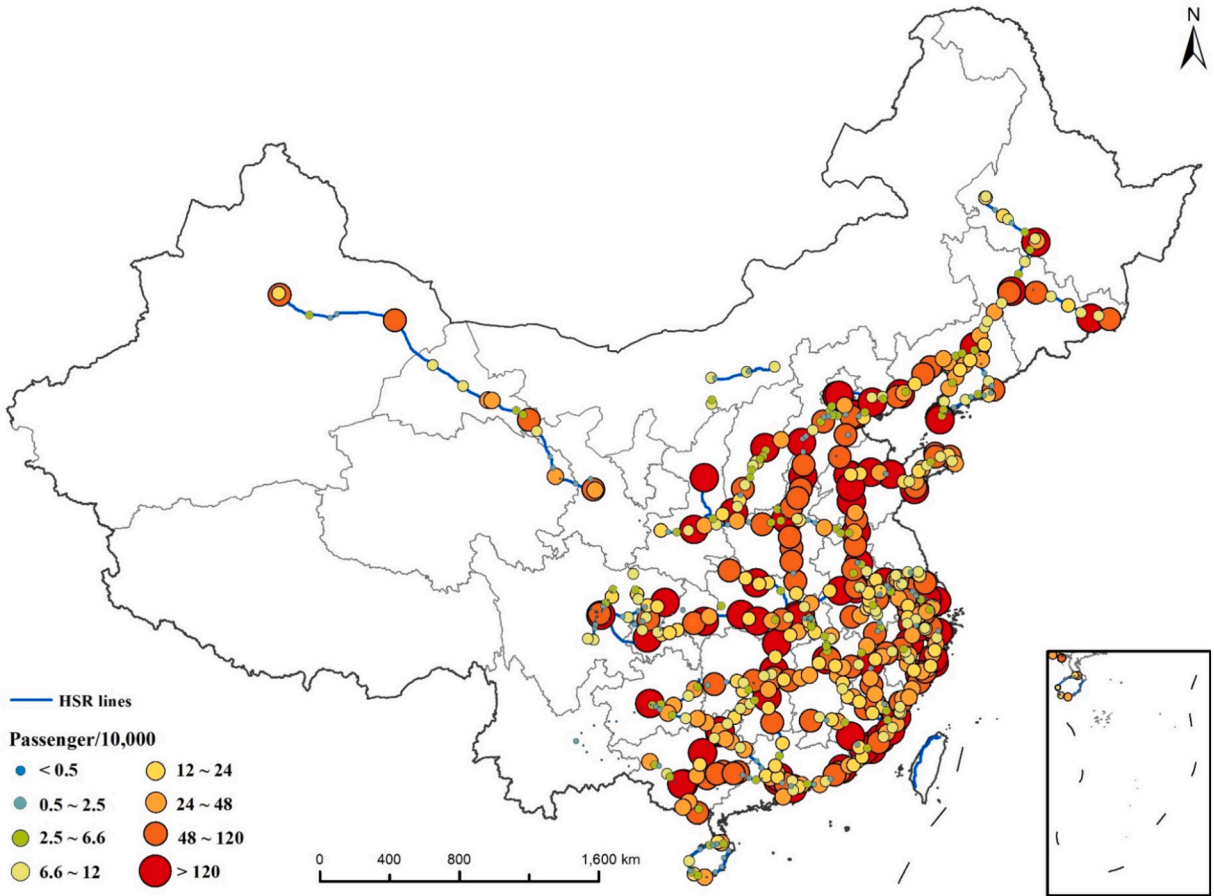


Fig. 1. Passenger numbers in 2016 (travel time > 2 h).

cities. Many of them also have airports and serve as key hubs within regional or even national intercity transportation networks.

3.1.2. Weather conditions

Daily weather data from 2014 to 2016 are obtained from the China Meteorological Data Service Centre, which is affiliated with the National Meteorological Information Centre.⁴ The dataset includes daily weather conditions from 2,420 basic and reference surface meteorological observation stations across mainland China, as presented in Fig. 2(a). It provides information on precipitation, wind speed, wind direction, temperature (maximum, minimum, and average), relative humidity, and atmospheric pressure. Since weather conditions such as thunderstorms, snowfall, and strong winds are major contributors to flight delays and cancellations (Gultepe, 2023), daily cumulative precipitation is used in the baseline analysis, whereas wind speed is used in the robustness checks. To generate nationwide daily weather grids, we follow the literature and apply spline interpolation on the basis of the geographic locations of meteorological stations (Samanta et al., 2012). Additionally, daily weather variables at the station level are constructed using the geographical locations of airports and HSR stations, as shown in Fig. 2 (b).

Fig. 3(a) and (b) show the frequency distributions of the daily cumulative precipitation and daily average wind speed at the station-day level from 2014 to 2016. As shown, the frequency distribution of precipitation exhibits a tailing pattern as the rainfall intensity increases, whereas the frequency distribution of the wind speed follows an inverted U-shaped pattern.

3.2. Empirical strategy and data summary

Owing to the unavailability of daily airport-level passenger flow data, we use a difference strategy to estimate HSR-air substitution under bad weather conditions. Specifically, we divide all HSR stations into two groups: the treatment group consists of stations with airports within their service area, whereas the control group includes stations without airports in their service area. By comparing the differences in passenger flow changes between the two groups of stations under good and bad weather conditions, we can accurately

⁴ For more details, please see <https://data.cma.cn/data/detail/dataCode/A.0012.0001.html>.

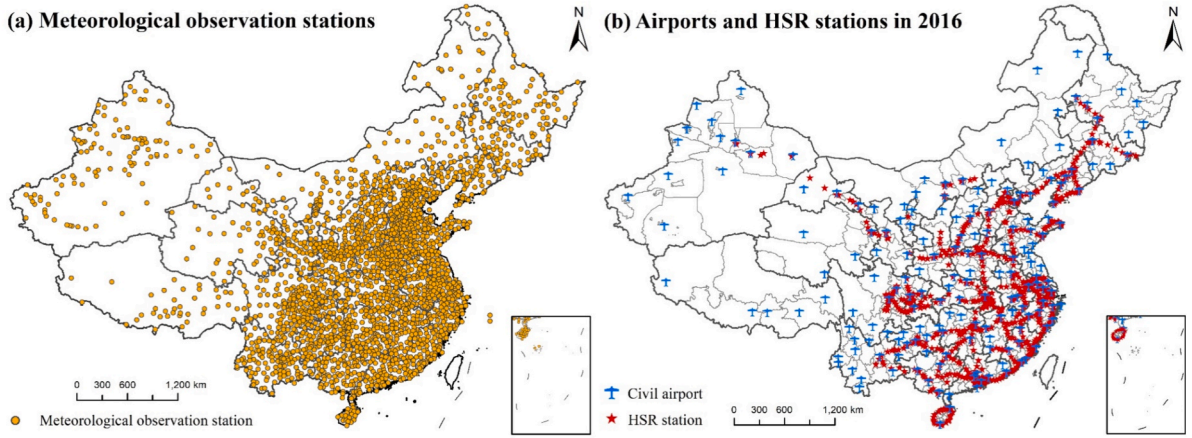


Fig. 2. Distribution of meteorological and transport stations, 2016.

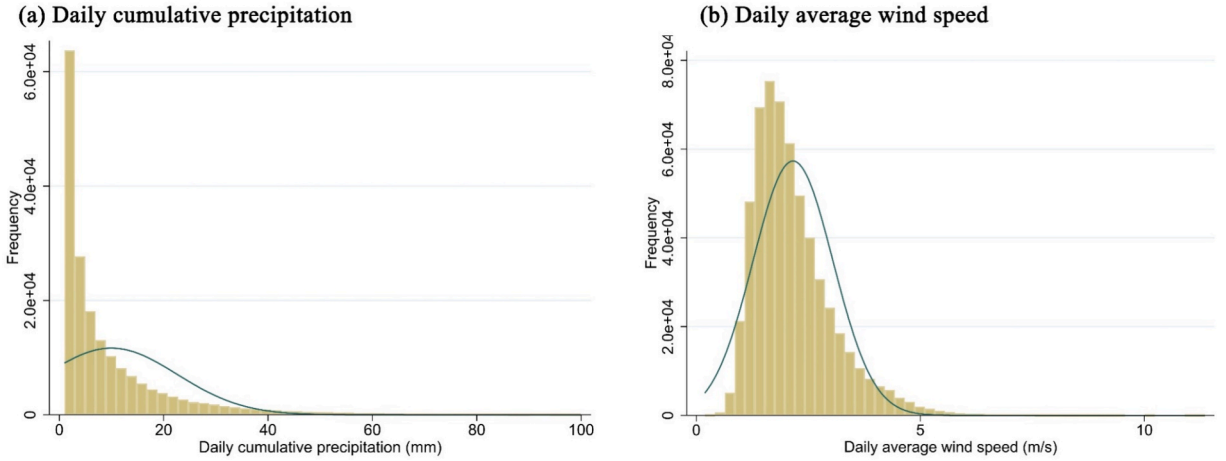


Fig. 3. Frequency distribution of precipitation and average wind, 2014–2016 Notes: Since non-rainy days constitute the majority, Fig. 3(a) presents only days with daily cumulative precipitation between 1 mm and 100 mm.

estimate the number of passengers shifting from air travel to HSR travel during bad weather events. Specifically, our baseline econometric model with high-dimensional fixed effects is formulated as follows:

$$\ln(\text{Passenger}_{icd}) = \alpha_0 + \alpha_1 \text{Badweather}_{icd} \times \text{Airport}_{icd} + \alpha_2 \text{Badweather}_{icd} + \alpha_3 \text{Airport}_{icd} + \gamma_k + \theta_d + \delta_i + \tau_{cm} + \varepsilon_{icd} \quad (1)$$

where $\ln(\text{Passenger}_{icd})$ represents the natural logarithm of the number of departure passengers whose travel time exceeds 2 h at HSR station i in city c on day d . Here, we do not count the passenger flow for short-distance travel with an HSR travel time of less than 2 h because there are few competitive substitutions between HSR and air transport in these short-distance travels (Lawrence et al., 2019).

The term Badweather_{icd} is a day-varying dummy variable used to measure whether station i experiences bad weather on day d . Bad weather in this study is defined by a set of criteria. In the baseline estimation, Badweather_{icd} equals 1 if the daily cumulative precipitation at the station exceeds 10 mm⁶ and 0 otherwise. According to the precipitation intensity classification standards issued by the China National Meteorological Administration, a daily cumulative precipitation of 10 mm is considered moderate rain, and more than 25 mm is classified as heavy rain.⁷ In Sections 5.1 and 4.2, we test other precipitation thresholds and the wind speed as the

⁵ An important reason for not using airport-based weather as the primary measure for Badweather_{icd} is that the vast majority (76%) of HSR stations do not have an airport within their service area. For robustness, we reconstruct the Badweather_{icd} variable using daily airport weather data for HSR stations that do have airports in their service areas (see Section 4.2.3).

⁶ As described in Section 3.1.2, we follow the literature and apply spline interpolation to the geographic coordinates of 2,420 weather stations to generate a nationwide daily weather grid. Using the location of each transportation station, we then obtain the corresponding daily weather conditions.

⁷ For more details, please see https://www.gov.cn/ztl/2008tffy/content_1113935.htm.

measure of bad weather, respectively.⁸

Airport_{icd} is a binary variable indicating whether an airport is operating within the service area of HSR station i on day d . In the baseline analysis, we define the service area of intercity transportation stations based on county-level administrative boundaries. Specifically, for stations located in central urban districts, the service area encompasses the entire prefecture (including both urban districts and counties), whereas for stations located in peripheral counties, the service area is defined as the county.⁹ In the robustness checks, we use alternative methods to delineate the service areas of the stations.

We further control for high-dimensional fixed effects. γ_k represents the fixed effects for the day of the week (Monday to Sunday), which control for systematic variations in intercity travel demand between weekdays and weekends, such as commuting-related travel versus leisure-oriented trips. θ_d denotes the fixed effects for the week of the year on day d , which flexibly capture intra-annual seasonal fluctuations in travel demand, including major holiday travel peaks (e.g., summer vacations and winter holidays), as well as broader cyclical patterns within the year. δ_i is the fixed effects for HSR station i . It capture a wide range of time-invariant station-specific characteristics, including the station's geographic location within the city, its position and hierarchy within the HSR network, and its overall scale of operation (e.g., station size, baseline service capacity, and long-term service frequency). τ_{cm} refers to the county-by-month fixed effects for departure counties to control for time-varying local demand conditions. These include variations in local economic activity, tourism attractiveness, and population mobility, as well as seasonal fluctuations such as holidays, climate patterns, and recurring demand cycles.¹⁰ ϵ_{icd} indicates the random error clustered at the station level. α_1 is the coefficient of interest.

It is worth noting that China's HSR network continued to expand during the sample period (2014–2016), with new lines and stations gradually entering operation. Our station-day empirical framework explicitly accommodates this dynamic process. Specifically, passenger flow observations for a given HSR station are included in the regressions only after the station becomes operational, resulting in an unbalanced panel structure. Newly opened stations therefore do not mechanically inflate substitution effects but contribute to identification only when they are actively serving passengers. Moreover, the inclusion of high-dimensional fixed effects at the station and time levels absorbs time-invariant station characteristics and common temporal shocks, ensuring that the estimated HSR-air substitution effects are not confounded by contemporaneous network expansion.

Given this rich set of fixed effects, our identification strategy effectively controls for potential variations in HSR demand, such as reduced willingness among passengers and mode shifts from driving. Therefore, α_1 captures the net substitution effect of HSR travel for air travel. A significant and positive α_1 indicates that, compared with normal weather, passenger flow at HSR stations with airports within their service area significantly increases under bad weather conditions. In other words, HSR-air substitution increases significantly during bad weather, with more passengers shifting from air travel to HSR travel.

The summary statistics are shown in Table 2. We keep all HSR travel records with travel times exceeding 2 h and aggregate them at the station-day level. During the study period from 2014 to 2016, the number of HSR stations gradually increased as new stations opened. Specifically, the numbers of operational HSR stations during these three years are 454, 664, and 774, respectively. In total, we obtain 574,994 observations at the station-day level. On average, a station has 2,900 passengers per day with travel times exceeding 2 h, but there are substantial variations across stations and days. In terms of travel distance, the number of passengers decreases as travel time increases. The proportions of passengers with travel times of 2–4 h, 4–7 h, and more than 7 h are 58.25%, 33.38%, and 8.37%, respectively.

4. Baseline results

4.1. Baseline estimations

Table 3 lists the regression results of Eq. (1). Column (1) shows the baseline estimation results for total passenger trips exceeding 2 h. In columns (2)–(4), we further decompose the total number of passenger trips into those with durations of 2–4 h, 4–7 h, and more than 7 h. We control for day-of-week fixed effects, week-of-year fixed effects, station fixed effects, and county-by-month fixed effects in all columns. The standard errors are clustered at the station level.

The coefficient of the interaction term (**Badweather** $\times$ **Airport**) is positive and statistically significant at the 95% confidence level, indicating that the increase in HSR competitive advantage under bad weather conditions leads to a significant increase in passenger flow. According to the results in column (1), compared with normal weather, HSR stations with airports within their service area experience a 2% higher passenger flow in bad weather than stations without airports. Importantly, this elasticity reflects a substantial volume of air-HSR substitution. On the basis of the daily passenger volumes at HSR stations from 2014 to 2016, we estimate that approximately 1.83 million HSR trips exceeding 2 h were shifted from air travel to HSR travel during bad weather over the three years. This finding highlights the critical role of the HSR system in enhancing the resilience of intercity travel under bad weather conditions.

⁸ Thunderstorms, dense fog, and other bad weather events can also significantly affect air operations. However, consistent nationwide daily data for these conditions are not available.

⁹ In China, a prefecture typically consists of dozens of central urban districts and surrounding counties. The distinction between districts and counties lies in the population density and intensity of economic activity. Districts are generally economically developed and densely populated areas located in the central part of a prefecture, whereas counties are often situated in suburban and peripheral areas.

¹⁰ We further conduct robustness checks by explicitly controlling for HSR train frequency and market access. Specifically, we incorporate HSR train frequency at the station-year level and market access at the county-year level as additional control variables. The estimation results remain highly stable and consistent with the baseline findings. Detailed results are available upon request.

Table 2
Variable definitions and summary statistics.

Variable	Description	Mean	Std. Dev.	Min	Max
Panel A: Daily passenger number at HSR station level					
Passenger	Daily number of HSR passengers for trips lasting over 2 h	2,900.35	8,190.52	1	196,012
Passenger_2-4	Daily number of HSR passengers for trips lasting 2–4 h	1,689.47	4,286.84	0	64,086
Passenger_4-7	Daily number of HSR passengers for trips lasting 4–7 h	968.14	3,505.84	0	102,443
Passenger_7more	Daily number of HSR passengers for trips lasting over 7 h	242.73	1,001.81	0	38,420
Panel B: Independent variables					
Badweather	Dummy, 1 for HSR station-day under bad weather, 0 otherwise	0.10	0.30	0	1
Airport	Dummy, 1 for HSR station with airport within its service area, 0 otherwise	0.24	0.43	0	1

Notes: All the statistics are reported for each station-day during the sample period of 2014–2016. The number of observations is 574, 994. The baseline definition of **Badweather** is based on the threshold of cumulative precipitation exceeding 10 mm; a set of criteria is used in [Sections 4.2 and 5.1](#).

Table 3
Baseline estimations.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2–4 h	4–7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.02** (0.009)	0.017* (0.009)	0.025*** (0.009)	0.009 (0.011)
Badweather	−0.018*** (0.004)	−0.018*** (0.004)	−0.018*** (0.005)	−0.011* (0.006)
Airport	−0.013 (0.158)	−0.005 (0.186)	−0.018 (0.108)	0.130 (0.080)
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
R ²	0.954	0.947	0.940	0.903
Observation	574,994	574,994	574,990	574,991

Notes: This table presents the HSR-air substitution under bad weather conditions. The variable **Badweather** measures weather conditions at HSR stations rather than at airports since many HSR stations do not have airports within their service areas. As a robustness check, we also use weather conditions at nearby airports to construct **Badweather**. The results remain consistent and are reported in [Section 4.2.3](#). For each column, the dependent variables are the natural logarithm of the number of HSR passengers at the station-day level. Columns (2)–(4) report the results for HSR trips with durations of 2–4 h, 4–7 h, and more than 7 h, respectively. The standard errors are reported in parentheses and clustered at the HSR station level. *, **, and *** refers to the statistically significant at 10%, 5%, and 1%, respectively.

and provides empirical support for [Hypothesis 1](#).

Specifically, the results in columns (2)–(4) show that the largest increase in passenger flow occurs for trips lasting 4–7 h, followed by trips lasting 2–4 h. While there is also an increase in passenger flow for trips exceeding 7 h, it is not statistically significant. We further estimate a series of regressions by grouping HSR trips into one-hour intervals. The results remain consistent: for trips shorter than 7 h, the interaction term remains significantly positive across all groups. However, for trips longer than 7 h, the substitution effect becomes statistically insignificant.¹¹ Between 2014 and 2016, HSR trips lasting 4–7 h in China typically correspond to geographic distances of 1,000–1,750 km, which is also the distance range where HSR-air competition and substitution are most intensive ([Lawrence et al., 2019; Ma et al., 2019; Zhang et al., 2019](#)). In summary, these heterogeneous patterns by travel distance provide strong evidence for [Hypothesis 2](#).

4.2. Robustness checks

4.2.1. Measure of bad weather

Although daily cumulative precipitation is a suitable indicator of bad weather, other potential measures exist. To test the robustness

¹¹ Due to space constraints, the detailed regression results are available from the authors upon request.

of the baseline findings, we use daily average wind speed as an alternative measure for bad weather. In accordance with the Civil Aviation Administration of China's regulations on civil aviation operations, we select daily average wind speeds greater than 5.5 m/s and greater than 8 m/s as alternative indicators for bad weather¹² and rerun the regressions by employing Eq. (1). The results are presented in Table A1 of Appendix A.

The organization of Table A1 is similar to that of Table 3. In columns (1)–(4), the dependent variables are the natural logarithms of the number of passengers for trips lasting more than 2 h, 2–4 h, 4–7 h, and more than 7 h, respectively. Panels A and B show the regression results where daily average wind speeds of 5.5 m/s and 8 m/s are used as thresholds for bad weather. When these alternative variables replace the baseline bad weather variable, the interaction term (*Badweather* × *Airport*) remains significantly positive in column (1) for both Panels A and B. When decomposed by travel distance, this significant positive effect is driven primarily by HSR trips lasting 4–7 h. These results are consistent with the baseline findings in Table 3.

4.2.2. Fixed effects

We test the robustness of the baseline findings by using an alternative set of high-dimensional fixed effects. Specifically, we replace the day-of-week fixed effects and week-of-year fixed effects from Table 3 with more stringent day fixed effects. The results of the re-regression are listed in Table A2 of Appendix A. The statistical significance, sign, and magnitude of the interaction term coefficients are consistent with those from the baseline regressions in Table 3.

4.2.3. Local weather conditions

In the baseline estimations, the *Badweather* variable measures the daily weather conditions at the HSR stations. In this subsection, we construct the *Badweather* variable using the daily weather conditions at airports and re-estimate the regression using Eq. (1). Specifically, if the airport falls within the service area of a HSR station (as defined in Section 3.2), we use the weather conditions observed at the corresponding airport to define the *Badweather* variable. If no airport falls within the HSR station's service range, we retain the HSR station's weather as the measure. Following the baseline estimation, *Badweather* equals 1 if the daily cumulative precipitation exceeds 10 mm and 0 otherwise. The regression results, shown in Table A3 of Appendix A, confirm that the baseline findings remain highly robust.

4.2.4. Spatial matching between HSR stations and airports

In the baseline regressions, we define the service areas of HSR stations and airports using the administrative boundaries of urban areas and counties and perform spatial matching between HSR stations and airports. However, airports that serve multiple cities simultaneously may affect the spatial matching process. Here, we manually identify airports that serve multiple cities and match them with the HSR stations of multiple cities. Specifically, we collect information on airports known to serve two or more cities (as listed in Table A4 of Appendix A) and match each of these airports to the HSR stations in all cities they serve. For example, Xi'an-Xianyang International Airport provides air travel services to residents in both Xi'an and Xianyang. Thus, we match the airport with the HSR stations in both cities. When bad weather conditions affect this airport, the nearby HSR stations in both cities are considered potential receivers of passengers diverted from air travel.

We then re-construct the *Airport* variable on the basis of this refined multi-city matching scheme and rerun the regressions. The results are listed in Table A5 of Appendix A, where the coefficients are highly consistent with the baseline estimations.

4.2.5. Excluding extreme weather conditions that disrupt HSR operations

While the HSR system is significantly more resilient to bad weather conditions than the aviation system is, it does not remain fully operational under all weather scenarios. According to the Grade of High Impact Weather Conditions for High-Speed Railway Operation (QX/T 334–2016)¹³ issued by the China Meteorological Administration, extreme rainfall events with hourly precipitation exceeding 50 mm or daily precipitation exceeding 100 mm can severely disrupt HSR operations. Under such conditions, train speeds must not exceed 80 km/h, whereas flood-prone sections are further restricted to a maximum of 45 km/h, with possible service suspensions in extreme cases. To account for these operational constraints, we exclude observations where daily precipitation exceeds 100 mm and rerun Eq. (1). The results, which are summarized in Table A6 of Appendix A, remain highly consistent with our baseline findings.

4.2.6. Alternative indicator for airport

We re-run the regressions using an alternative measure of the *Airport* variable in Eq. (1). Specifically, we use the logarithm of the annual number of flights at each airport-year as the alternative measure of air service. The empirical results, as shown in Table A7 of Appendix A, are consistent with the baseline regressions. The coefficients of the interaction term are significantly positive in the regressions for travel distance over 2 h, 2–4 h, and 4–7 h.

In summary, the results of the robustness checks listed in Appendix A demonstrate that the baseline findings of this study are not biased by model setting or variable measurement.

¹² According to the minimum operating standards for civil aviation operations established by the Civil Aviation Administration of China (https://www.gov.cn/gongbao/content/2002/content_61982.htm), aircraft can still operate at wind speeds of 5.5 m/s and 8 m/s. However, it should be noted that these values refer to the average wind speed over the entire day. Under such average wind conditions, aircraft operations can still be significantly affected.

¹³ For more details, please see https://www.cma.gov.cn/zfxgkg/gknr/flgbz/bz/202209/t20220921_5098278.html.

5. Heterogeneity analysis

If the coefficients identified in the baseline regressions are driven by passengers switching from air travel to HSR during bad weather conditions, we would expect to observe two specific patterns. First, the substitution effect should become more pronounced under heavier rainfall, as more severe weather leads to longer flight delays and higher cancellation rates, as discussed in [Hypothesis 3](#). Second, the effect should be stronger in cities where the accessibility between airports and HSR stations is higher, as described in [Hypothesis 4](#). Therefore, this section empirically tests these two sources of heterogeneity to provide additional evidence for the HSR-air substitution effect during bad weather events.

5.1. Rainfall intensity

HSR-air competition and substitution are influenced by the severity of weather conditions. As discussed in [Hypothesis 3](#), HSR systems are significantly more resilient and tolerant to bad weather than aviation systems are. As weather conditions worsen, the probability of flight delays or cancellations increases, and the duration of delays becomes longer ([Chen et al., 2021](#)). Consequently, the competitive advantage of HSR increases with the severity of bad weather. To investigate the volume of passengers shifting from air travel to HSR travel under varying levels of bad weather, we test a series of weather thresholds. Specifically, we define bad weather thresholds on the basis of daily cumulative precipitation levels of 5 mm, 10 mm, 15 mm, 20 mm, 25 mm, and 30 mm, and re-run the regressions, respectively.

[Fig. 4](#) presents the magnitudes of the interaction term coefficients and their corresponding 95% confidence intervals for each precipitation threshold. In these regression analyses, the dependent variable is the logarithm of daily trip counts at HSR stations for the 4–7 h travel distance.¹⁴ It is evident that as the severity of bad weather increases, the magnitude of the regression coefficients increases gradually, from 1.77% to 5.69%. This finding indicates that [Hypothesis 3](#) holds. As the weather worsens, the number of passengers shifting from air travel to HSR travel increases.

The pattern shown in [Fig. 4](#) is related to the differential sensitivity of air travel and HSR travel to bad weather conditions and is consistent with the literature ([Chen et al., 2021](#)). The resilience of intercity transport infrastructure, particularly air travel and HSR, under unexpected shocks has been discussed in the literature ([Li et al., 2024](#); [Zhou, 2024](#)). It is widely acknowledged that HSR is less affected by bad weather than air travel, as discussed in [Section 2](#). In the decision-making process of travelers, the expected (time) loss from air travel during bad weather is the product of the probability of flight delays and the corresponding delay duration. Both of these factors are significantly positively correlated with the severity of the weather. As weather conditions worsen, the expected (time) loss from air travel increases, whereas the competitive advantage of HSR gradually increases.

5.2. Intra-city accessibility between HSR stations and airports

The intra-city distance between HSR stations and airports is another key factor influencing HSR-air substitution under bad weather conditions, as described in [Hypothesis 4](#). Previous studies have shown that poor intra-city accessibility between rail stations and airports is a significant reason for low air-rail intermodality performance ([Albalade et al., 2015](#)). Similarly, intra-city accessibility between airports and HSR stations plays a critical role in passengers' decisions to switch from air travel to HSR under bad weather conditions ([Wu and Han, 2022](#)), especially when scheduled flights are significantly delayed or canceled. In such scenarios, passengers with essential travel face the choice of either continuing to wait for their flight or switching to HSR. In this context, the accessibility between airports and HSR stations significantly affects passengers' decisions, as it adds to the time cost of switching modes of transport.

To examine the impact of intra-city accessibility, we categorize HSR stations with airports within their service areas into two distinct subgroups based on the driving time between each airport-HSR-station pair: those with a driving time of less than 60 min and those with a driving time greater than 60 min. The control group for both subgroups consists of all HSR stations that did not have an airport in their service areas by the end of 2016.

The results of the heterogeneity analysis by employing Eq. (1) are presented in [Table 4](#). Panel A presents the regression results for the subgroup of HSR stations with a driving time to airports of less than 60 min, while Panel B shows the results for those with a driving time of more than 60 min. In column (1) of Panel A, compared with good weather, the number of passengers (for trips longer than 2 h) at HSR stations with airports increases by 2.4% during bad weather, and the result is statistically significant at the 95% level. When decomposed by travel duration, the coefficient for trips lasting 4–7 h reaches a maximum of 2.78% and is statistically significant at the 99% confidence level, as shown in the last three columns of Panel A. In contrast, the estimates in Panel B show that for HSR stations with a driving time to the airport of over 60 min, the coefficients of the interaction term are statistically insignificant. This pattern holds for all trip durations, regardless of whether the dependent variable is the number of trips exceeding 2 h, 2–4 h, or 4–7 h.

The findings in [Table 4](#) highlight that intra-city accessibility is a crucial factor in passengers' decisions to shift from air travel to HSR travel during bad weather. This pattern is consistent with existing studies, which suggest that the intra-city location and accessibility of HSR stations significantly influence demand for HSR travel ([Diao et al., 2017](#); [Zheng et al., 2019](#); [Zheng and Wu, 2024](#)).

¹⁴ The 4–7 h HSR travel distance is the range where HSR-air competition and substitution is most intensive. We also perform a test using the logarithm of daily trip counts for HSR stations with travel times exceeding 2 h as the dependent variable. The results are similar with those in [Fig. 4](#) and are available upon request.

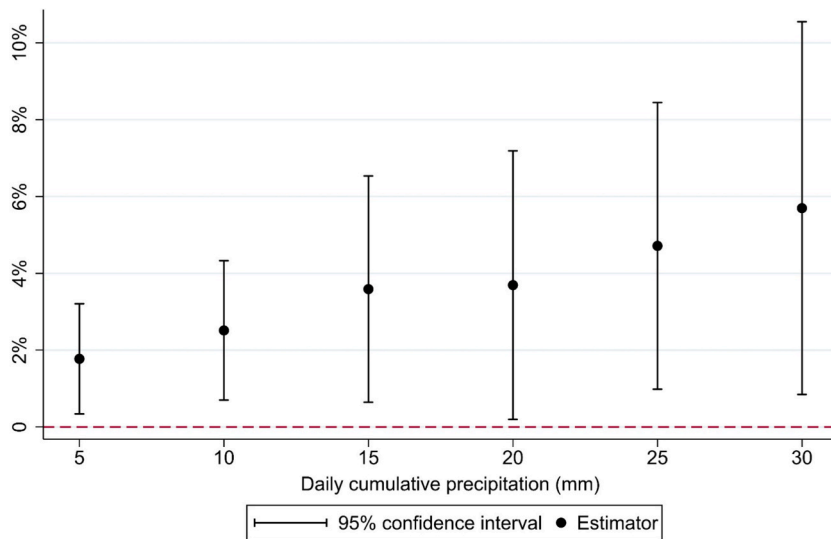


Fig. 4. Heterogeneities across different rainfall thresholds.

Table 4
Heterogeneities by intra-city accessibility.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Panel A: Driving time < 60 min				
Badweather × Airport	0.024** (0.011)	0.019* (0.011)	0.028*** (0.011)	0.015 (0.012)
Number of stations	734	734	734	734
R ²	0.952	0.945	0.939	0.902
Observation	552,751	552,751	552,747	552,748
Panel B: Driving time ≥ 60 min				
Badweather × Airport	0.011 (0.012)	0.013 (0.010)	0.014 (0.012)	−0.017 (0.021)
Number of stations	611	611	611	611
R ²	0.951	0.948	0.933	0.885
Observation	461,021	461,021	461,018	461,019
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes

Notes: For each column, the dependent variables are the natural logarithm of the number of HSR passengers at the station-day level. The standard errors are reported in parentheses and clustered at the HSR station level. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

6. Further discussions

6.1. Calendar effect

The necessity of travel significantly influences the shift from air travel to HSR travel under bad weather conditions. Since we do not have access to the specific purpose of each HSR trip, we classify travel dates into three categories: workdays (excluding weekends and public holidays), short holidays (weekends and public holidays lasting 2–3 days¹⁵), and long holidays (National Day and Spring Festival holidays lasting 7 days). These categories differ significantly in terms of travel purpose and necessity. First, intercity travel necessity is highest on weekdays, as business-related trips are work-driven and largely mandatory. Second, long-holiday travel also

¹⁵ In China, the public holidays lasting 2–3 days include Qingming Festival, Labor Day, Dragon Boat Festival, Mid-Autumn Festival, and New Year's Day.

exhibits strong necessity, albeit for different reasons. During the National Day holiday, high pre-booking costs for hotels and tourist attractions, along with extended vacation durations of seven days or more, make intercity travel relatively inflexible. During Chinese New Year, the demand for returning hometowns to reunite with family is exceptionally strong, making travel nearly unavoidable. Finally, short holidays, such as weekends and brief public holidays, are associated with the lowest travel necessity. Travel during these periods is typically for short leisure trips that are flexible to nearby destinations, making travel largely discretionary. Under bad weather conditions, travelers are more likely to cancel or postpone their trips than to switch to alternative transportation modes.

On the basis of subsamples from these three categories, we examine the calendar effect of HSR-air substitution behaviors under bad weather conditions by running regressions in Eq. (1). Table 5 summarizes the results, with panels A, B, and C presenting findings for workdays, short holidays, and long holidays, respectively. The structure of Table 5 is consistent with that of Table 3. In the first column, the coefficients of the interaction term are positive across all three panels, aligning with the baseline findings. However, Column (3) reveals notable calendar heterogeneity. The coefficient is statistically significant at the 95% confidence level for both long holidays and workdays, with the coefficient magnitude for long holidays being significantly larger than that for workdays. In contrast, the coefficient during short holidays is not statistically significant. This pattern suggests that HSR-air substitution effect under bad weather conditions is more pronounced for intercity travel with greater necessity.

6.2. Evidence at the station-pair-day level

Previous studies have extensively examined the impact of weather on air travel, highlighting that flight operations are influenced by natural conditions at both departure and destination airports (Chen et al., 2021; Dai et al., 2024). The previous sections provide empirical evidence at the station-day level. To more accurately capture HSR-air substitution under bad weather conditions, we extend the regression analysis to the station-pair-day level. By aggregating the trip-level data, we obtain the daily station-pair-level HSR passenger numbers.

Specifically, we estimate the following econometric model with high-dimensional fixed effects:

$$\begin{aligned} \ln(\text{Passenger}_{ijd}) = & \beta_0 + \beta_1 \text{Badweather_OD}_{ijd} \\ & \times \text{Airline}_{ijd} + \beta_2 \text{Badweather_O}_{id} + \beta_3 \text{Badweather_D}_{jd} + \beta_4 \text{Airline}_{ijd} + \gamma_{im} + \theta_{jm} + \delta_{ij} + \varepsilon_{icd} \nu \end{aligned} \quad (2)$$

where $\ln(\text{Passenger}_{ijd})$ represents the natural logarithm of the number of HSR passengers traveling between HSR station i and station j on day d . $\text{Badweather_OD}_{ijd}$ is a day-varying binary variable that equals one if the daily cumulative precipitation at either departure HSR station i or destination station j exceeds 10 mm, and zero otherwise.¹⁶ Airline_{ijd} is a dummy indicating whether there is an active commercial air route between the city-pair where HSR stations i and j are located. Since daily air route data are unavailable, we use a proxy measure. Airline_{ijd} takes a value of one if both cities where stations i and j are located have airports with more than 100 weekly flights; otherwise, it is set to zero.¹⁷ Badweather_O_{id} and Badweather_D_{jd} are dummies that capture the weather conditions at departure station i and destination station j on day d , respectively. Again, we further control for high-dimensional fixed effects. γ_{im} and θ_{jm} are county-by-month fixed effects for departure and destination counties, respectively. δ_{ij} represents station-pair fixed effects.

The regression results at the station-pair-day level are summarized in Table 6. Column (1) presents the regression results for station-pairs with HSR travel times exceeding 2 h, after controlling for county-by-month fixed effects for both departure and destination counties. In Column (2), we further tighten the fixed effects to station-month fixed effects for both departure and destination stations.

Across the two columns in Table 6, the coefficients for the interaction term are positive and statistically significant at the 99% confidence level, indicating that under bad weather conditions, station-pairs with competitive air routes experience a significant increase in passenger volume. To provide a concrete interpretation, consider the Guangzhou-Zhengzhou corridor. The straight-line distance between the two cities is approximately 1,303 km, and the fastest HSR services complete the trip in about five hours. Under normal weather conditions, aviation retains a time advantage and accounts for roughly 59% of the passenger market on this route.¹⁸ However, when adverse weather disrupts flight operations at the departure airport, the reliability advantage of air travel diminishes sharply. In contrast, HSR services on this corridor remain largely stable, offering predictable travel times and high service frequency. Faced with increased uncertainty and expected delays in air travel, many passengers switch to HSR, leading to a significant increase in HSR passenger volumes between Guangzhou and Zhengzhou on bad-weather days.

Additionally, the coefficient of Badweather_O (bad weather at the departure station) is notably larger than that of Badweather_D (bad weather at the destination station), suggesting a myopia effect in travel decision-making. This aligns with findings in the literature, which suggest that decision-makers tend to overemphasize current costs and benefits while underestimating future or uncertain ones (Laverty, 1996). In the context of intercity travel, passengers appear to be more sensitive to weather conditions at their departure location than at their destination.

¹⁶ Following the tests in Section 4.2, we conduct a series of robustness checks on the measurement of bad weather, and the empirical results remains consistent. These results are available upon request.

¹⁷ We also test alternative measures, and the results remain robust. For more details, see Appendix B.

¹⁸ Data are from the Civil Aviation Data Analysis Data Hub (<https://www.cadas.com.cn/>).

Table 5
Calendar effect.

	Ln (Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Panel A: workdays				
Badweather × Airport	0.018* (0.010)	0.016 (0.010)	0.021** (0.009)	0.007 (0.011)
R ²	0.956	0.949	0.942	0.906
Observation	394,101	394,101	394,099	394,098
Panel B: short holidays (2–3 days)				
Badweather × Airport	0.024 (0.018)	0.016 (0.017)	0.022 (0.020)	0.007 (0.019)
R ²	0.954	0.948	0.941	0.904
Observation	158,938	158,938	158,936	158,938
Panel C: long holidays (7 days)				
Badweather × Airport	0.003 (0.035)	0.001 (0.033)	0.105** (0.042)	−0.023 (0.040)
R ²	0.947	0.941	0.927	0.882
Observation	72,194	72,194	72,194	72,194
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760

Notes: The standard errors are reported in parentheses and clustered at the HSR station level. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table 6
Evidence at the station-pair-day level.

	Ln (Passenger)	
	(1)	(2)
Badweather × Airline	0.007*** (0.002)	0.007*** (0.002)
Badweather_O	−0.01*** (0.001)	−0.01*** (0.001)
Badweather_D	−0.003*** (0.001)	−0.003*** (0.001)
County_O-month fixed effects	Yes	No
County_D-month fixed effects	Yes	No
Station_O-month fixed effects	No	Yes
Station_D-month fixed effects	No	Yes
Station-pair fixed effects	Yes	Yes
Number of station pair	24,160	24,158
R ²	0.899	0.901
Observation	5,000,950	5,000,941

Notes: Each unit represents an observation at the station-pair-day level. The regression sample includes HSR station-pairs with travel times greater than 2 h. In Column (1), departure county × month fixed effects, destination county × month fixed effects, and station-pair fixed effects are controlled, whereas departure station × month fixed effects, destination station × month fixed effects, and station-pair fixed effects are controlled in Column (2). The standard errors are reported in parentheses and clustered at the HSR station level. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

6.3. Welfare analysis

This subsection provides a back-of-the-envelope estimation of the welfare gains resulting from travel time savings enabled by the HSR system under bad weather conditions. The conceptual formula for the estimation is as follows:

$$\text{Annual increased welfare} = \sum_{i=1}^m \sum_{d=1}^n \text{Time_saving}_{id} \times \text{Passenger}_{id} \times \text{Wage}_i \quad (3)$$

In Eq. (3), the parameter m represents the total number of cities with HSR stations, whereas n denotes the total number of days with bad weather in a year. *Time_saving_{id}* refers to the travel time savings experienced by travelers in city i on day d due to the high resilience of the HSR system under bad weather conditions. For this estimation, we assume that passengers switching from air travel to HSR travel behave as rational-economic people. In this setting, under bad weather conditions, the delay time associated with air travel is assumed to be equivalent to the time savings gained by switching to HSR travel.¹⁹ On the basis of the Airport Capacity and Delay report by the U.S. Federal Aviation Administration (FAA),²⁰ the Performance Review Report by EUROCONTROL,²¹ and annual operational data from Chinese airports,²² we obtain three sets of estimated delay times for Chinese airports under different weather conditions, as presented in Table C1 of Appendix C.

Passenger_{id} represents the number of travelers in city i on day d who switch from air travel to HSR travel due to bad weather. This value can be estimated using the daily passenger volume at HSR stations and the coefficients derived in Table 3. Based on daily HSR ridership data from 2014 to 2016, we estimate that approximately 616 thousand, 526 thousand, and 685 thousand passenger trips, shifted from air to HSR under bad weather conditions in 2014, 2015, and 2016, respectively.

Wage_i denotes the average wage of intercity travelers in city i and serves as a proxy for their time value. Because individual-level income data for travelers are unavailable, we use the annual average wage of the city in which the airport is located as a proxy. This assumption is intentionally conservative. In practice, air travelers typically have higher incomes and higher values of time than the city average. As a result, our welfare estimates should be interpreted as a lower bound of the true welfare gains generated by HSR-air substitution under bad weather conditions.

We summarize the welfare gains for intercity travelers enabled by the HSR system under different scenarios, as shown in Table 7. Compared with aviation systems, HSR systems are significantly less affected by bad weather. The availability of HSR provides intercity travelers with greater flexibility, serving as a valuable transportation option that substantially reduces travel time during bad weather conditions. These time savings translate into considerable welfare gains. Even during the 2014–2016 period, when China's HSR network was still in its early expansion phase, the estimated welfare gains (under conservative assumptions) ranged from 151.61 to 400.13 million RMB in these three years. As the HSR network continues to expand after 2016 and bad weather events become more frequent due to climate change, the welfare benefits of HSR systems for intercity travelers are expected to grow further.

While these estimates are small relative to the overall scale of HSR investment and operating revenues (Chang et al., 2026), they represent an additional and previously underexplored dimension of HSR benefits. In particular, they capture the value of enhanced travel reliability and reduced uncertainty under adverse weather conditions, which is not reflected in conventional measures such as fare revenues or aggregate economic impacts. As such, these welfare gains should be interpreted as a complementary component of the broader benefits of HSR systems, rather than a primary source of economic return.

7. Conclusions

In the context of increasingly frequent extreme weather events driven by climate change, this study contributes to the literature by integrating weather scenarios into the analysis of HSR-air substitution and transport resilience. Unlike previous studies that largely rely on theoretical models and case studies, our empirical analysis is based on nearly 3.7 billion actual HSR ticket records from 2014 to 2016 across China. This full-sample dataset allows for a comprehensive examination of how bad weather influences passenger behavior in intercity travel and promotes travel resilience. Our findings highlight the critical role of weather in shaping the substitution dynamics between HSR and air travel. Specifically, bad weather significantly enhances HSR-air substitutions. On average, during bad weather conditions, 2% of HSR passengers traveling over two hours can be attributed to air trips shifted to HSR. This elasticity means that approximately 1.83 million trips exceeding 2 h were shifted from air travel to HSR during bad weather from 2014 to 2016.

This study also examines several key factors that influence HSR-air substitution under bad weather conditions. First, substitution behaviors during bad weather are observed primarily on intercity HSR trips lasting 2–7 h. For journeys exceeding 7 h, travelers are more likely to continue waiting for air travel despite unfavorable weather conditions. Second, intra-city accessibility between HSR stations and airports significantly affects passenger switching behavior from air to HSR during bad weather. This finding is consistent with the literature (Wu and Han, 2022), which emphasizes that station accessibility is a crucial determinant of HSR-air substitution, given that intra-city travel time is an essential component of total travel time. Third, the number of passengers shifting from air to HSR increases with increasing severity of weather conditions. This is because the probability and duration of flight delays are strongly positively correlated with weather severity (FAA, 1983; Li et al., 2024; Zhou, 2024). Finally, this shift occurs primarily on weekdays and long holidays, with no significant effect during short holidays. This pattern reflects the greater travel necessity associated with workdays and long holidays.

These findings have several significant policy implications. First, our findings highlight the importance of enhancing the resilience

¹⁹ The delay time does not deduct the transfer time between airports and HSR stations for two main reasons. First, during bad weather, airlines usually notify travelers of flight delays or cancellations several hours in advance, so most mode switches occur before travelers reach the airport. Second, for travelers transferring from airports to HSR stations, the significantly shorter boarding time at HSR stations in China largely offsets the time spent on airport-to-HSR transfers.

²⁰ For more details, please see https://www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/22824.

²¹ For more details, please see <https://www.eurocontrol.int/publication/performance-review-report-prr-2015>.

²² For more details, please see <https://www.gov.cn/xinwen/2017-05/08/5191876/files/df3832ba451c4a7eb70b5f419f6ba672.pdf>.

Table 7
Welfare gains from 2014 to 2016 (million).

	Baseline	Lower-bound	Upper-bound
Estimation by total passengers	287.59	164.28	400.13
Estimation by passengers with different travel durations	265.33	151.61	369.35

Notes: According to the estimation results in Table 3, compared with HSR stations without nearby airports, stations with airports in their service areas experience a significant increase in passenger flows under bad weather conditions. For trips exceeding 2 h, an estimated 2.03% of total HSR passengers are shifted from air travel during adverse weather conditions. When further broken down by travel duration, the proportion of passengers shifting from air to HSR is 1.65% for 2–4 h trips, 2.52% for 4–7 h trips, and 0.86% for trips exceeding 7 h.

of intercity travel systems amid growing climate uncertainty. Policymakers should prioritize the expansion and improvement of HSR networks, particularly on routes with durations between 2 and 7 h, where the competitive advantage of HSR travel over air travel is most evident. Given the increasing frequency of extreme weather events, expanding HSR services in these corridors can increase the overall resilience of intercity travel systems. Moreover, improving HSR scheduling flexibility during severe weather could further strengthen its role as a reliable alternative.

Second, improving the intra-city accessibility of HSR stations is crucial. Investments in integrated transport solutions, such as enhanced metro links, dedicated shuttle services, and seamless transfer facilities, can reduce total travel time and make HSR travel a more attractive option. These measures are particularly important in cities with substantial air travel demand, where improving HSR accessibility can significantly increase passenger switching during bad weather.

Third, transportation authorities should develop contingency plans that leverage the resilience of HSR systems during adverse weather events. For example, during periods of predictable bad weather, such as the June–September thunderstorm season in much of China, HSR operators could increase the frequency of services and allocate additional seats on 2–7 h routes to accommodate higher demand. In the short term, HSR systems can also dynamically adjust service supply based on weather forecasts, such as reallocating rolling stock, increasing seat capacity, or introducing temporary additional services. More broadly, integrating weather forecasting into operational planning can help align transport supply with fluctuating demand under climate uncertainty, thereby improving both reliability and passenger welfare.

CRediT authorship contribution statement

Xinlei Hu: Writing – review & editing, Software, Resources, Funding acquisition, Formal analysis, Data curation. **Shuping Wu:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **Longfei Zheng:** Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

We acknowledge the financial support by the National Natural Science Foundation of China (42301185, 52302396, 72474019, 42471291). Declarations of interest: none. All errors remain ours.

Appendix A.: Robustness checks

Table A1

Robustness check: alternative measure of bad weather.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Panel A: Badweather (wind > 5.5 m/s)				
Badweather × Airport	0.090* (0.049)	0.101** (0.048)	0.085* (0.051)	0.073 (0.048)
R ²	0.954	0.947	0.94	0.903

Panel B: Badweather (wind > 8 m/s)

(continued on next page)

Table A1 (continued)

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.466** (0.208)	0.543** (0.228)	0.328* (0.197)	0.454 (0.325)
R ²	0.954	0.947	0.94	0.903
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
Observation	574,994	574,994	574,990	574,991

Notes: In Panel A, the threshold of bad weather is that the daily average wind speed at the station exceeds 5.5 m/s, and the threshold is 8 m/s in Panel B. The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table A2

Robustness check: alternative set of fixed effects.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.020** (0.009)	0.016* (0.009)	0.025*** (0.009)	0.009 (0.011)
Day fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
R ²	0.954	0.947	0.94	0.902
Observation	574,994	574,994	574,990	574,991

Notes: In this table, day fixed effects, HSR station fixed effects, and county-by-month fixed effects are controlled. The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table A3

Robustness check: weather conditions in airports.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.017** (0.008)	0.014 (0.009)	0.022*** (0.008)	0.012 (0.009)
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
R ²	0.954	0.947	0.94	0.903
Observation	574,994	574,994	574,990	574,991

Notes: In this table, we construct the *Badweather* variable using the daily weather conditions at airports. The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table A4

Airports serving multiple cities in China, 2016.

Airport name	Cities served	Opening year
--------------	---------------	--------------

(continued on next page)

Table A4 (continued)

Airport name	Cities served	Opening year
Xi'an-Xianyang International Airport	Xi'an, Xianyang	1991
Yangzhou-Taizhou International Airport	Yangzhou, Taizhou	2012
Jieyang-Chaoshan International Airport	Jieyang, Shantou, Chaozhou	2011
Jiayuguan-Jiuquan Airport	Jiayuguan, Jiuquan	2001
Wuxi-Shuofang Airport	Wuxi, Suzhou	2004
Tongren-Fenghuang Airport	Tongren, Xiangxi	2001

Notes: Opening year refers to the year the airport began civil aviation operations.

Table A5

Robustness check: alternative matching methods.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.027*** (0.010)	0.023** (0.010)	0.031*** (0.010)	0.010 (0.012)
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
R ²	0.954	0.947	0.94	0.903
Observation	574,994	574,994	574,990	574,991

Notes: The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table A6

Robustness check: dropping days with extreme weather conditions.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × Airport	0.019** (0.009)	0.014* (0.008)	0.023** (0.009)	0.009 (0.011)
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760
R ²	0.954	0.948	0.94	0.903
Observation	574,420	574,420	574,416	574,417

Notes: In this table, we exclude observations where daily precipitation exceeds 100 mm. The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Table A7

Robustness check: alternative measure of air service.

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
Badweather × ln(Flight)	0.003* (0.001)	0.002* (0.001)	0.003** (0.001)	0.0002 (0.001)
Day-of-week fixed effects	Yes	Yes	Yes	Yes
Week-of-year fixed effects	Yes	Yes	Yes	Yes
HSR station fixed effects	Yes	Yes	Yes	Yes
County-by-month fixed effects	Yes	Yes	Yes	Yes
Number of stations	760	760	760	760

(continued on next page)

Table A7 (continued)

	Dependent variable: ln(Passenger)			
	Total (>2 h)	2~4 h	4~7 h	>7 h
	(1)	(2)	(3)	(4)
R ²	0.954	0.947	0.94	0.903
Observation	574,994	574,994	574,990	574,991

Notes: The term **ln(Flight)** refers to the logarithm of the annual number of flights at each airport-year. The standard errors are reported in parentheses and clustered at HSR station levels. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Appendix B:. Robustness check at the station-pair-day level

Following the existing literature (Wu and Han, 2022), we collect air travel data for 581 trunk routes from the Civil Aviation Administration of China (CAAC), which records annual passenger volumes for major city pairs during 2014–2016. The 581 routes connect 80 cities in China. Using these data, we reconstruct a measure of air transport connectivity at the city-pair level and re-estimated Eq. (2) at the station-pair-day level. Specifically, **Airline_{ij,d}** is a dummy indicating whether a trunk air route exists between the cities where HSR stations *i* and *j* are located. The re-estimation results, reported in Table B1, remain robust.

Table B1
Robustness check at the station-pair-day level.

	Ln (Passenger)	
	(1)	(2)
Badweather × Airline	0.022*** (0.006)	0.022*** (0.006)
Badweather_O	−0.009*** (0.0008)	−0.009*** (0.0008)
Badweather_D	−0.0016* (0.0008)	−0.0016* (0.0008)
County_O-month fixed effects	Yes	No
County_D-month fixed effects	Yes	No
Station_O-month fixed effects	No	Yes
Station_D-month fixed effects	No	Yes
Station-pair fixed effects	Yes	Yes
Number of station pair	24,160	24,158
R ²	0.899	0.901
Observation	5,000,950	5,000,941

Notes: Each unit represents an observation at the station-pair-day level. The regression sample includes HSR station-pairs with travel times greater than 2 h. In Column (1), departure county × month fixed effects, destination county × month fixed effects, and station-pair fixed effects are controlled, whereas departure station × month fixed effects, destination station × month fixed effects, and station-pair fixed effects are controlled in Column (2). The standard errors are reported in parentheses and clustered at the HSR station level. *, **, and *** refers to the statistically significant at 10%, 5%, and 1% levels, respectively.

Appendix C:. Delay times for Chinese airports

Table C1
Delay times for Chinese airports under different weather conditions.

	Baseline	Lower-bound	Upper-bound
Moderate rainfall (daily cumulative precipitation: 10–25 mm)	50	30	75
Heavy rainfall (daily cumulative precipitation: >25 mm)	90	50	120

Notes: The unit is minute.

Data availability

Data will be made available on request.

References

- Albalade, D., Bel, G., Fageda, X., 2015. Competition and cooperation between high-speed rail and air transportation services in Europe. *J. Transp. Geogr.* 42, 166–174.
- Behrens, C., Pels, E., 2012. Intermodal competition in the london–paris passenger market: high-speed rail and air transport. *J. Urban Econ.* 71 (3), 278–288.
- Bešinović, N., 2020. Resilience in railway transport systems: a literature review and research agenda. *Transp. Res.* 40 (4), 457–478.
- Chang, Z., Diao, M., Jing, K., Zheng, L., 2026. High-speed rail and economic growth: a market access-based economic impact evaluation in China. *Transp. Res. Interdiscip. Perspect.* 36, 101835.
- Chang, Z., Zheng, L., Yang, T., Long, F., 2022. High-speed rail, new town development, and the spatial mismatch of land leases in China. *Land Use Policy* 115, 106014.
- Chen, Z., 2017. Impacts of high-speed rail on domestic air transportation in China. *J. Transp. Geogr.* 62, 184–196.
- Chen, Z., Wang, Y., Zhou, L., 2021. Predicting weather-induced delays of high-speed rail and aviation in China. *Transp. Policy* 101, 1–13.
- China Meteorological Administration (CMA). (2016). Grade of high impact weather conditions for high-speed railway operation. Accessed December, 10, 2024. (https://www.cma.gov.cn/zfxgk/gknr/flfgbz/bz/202209/t20220921_5098278.html).
- Civil Aviation Administration of China (CAAC). (2001). Order of the Civil Aviation Administration of China. Accessed December, 10, 2024. (https://www.gov.cn/gongbao/content/2002/content_61982.htm).
- Civil Aviation Administration of China (CAAC). (2007). General operating and flight rules for civil aviation. Accessed December, 10, 2024. ([https://fsop.caac.gov.cn/files/19-%E4%B8%80%E8%88%AC%E8%BF%90%E8%A1%8C%E5%92%8C%E9%A3%9E%E8%A1%8C%E8%A7%84%E5%88%99\(CCAR-91\).pdf](https://fsop.caac.gov.cn/files/19-%E4%B8%80%E8%88%AC%E8%BF%90%E8%A1%8C%E5%92%8C%E9%A3%9E%E8%A1%8C%E8%A7%84%E5%88%99(CCAR-91).pdf)).
- Clewlow, R.R., Sussman, J.M., Balakrishnan, H., 2012. Interaction of high-speed rail and aviation: exploring air-rail connectivity. *Transp. Res. Rec.* 2266 (1), 1–10.
- Climate Central. (2023). Climate change is disrupting air travel. Accessed January, 5, 2025. (<https://www.climatecentral.org/climate-matters/climate-change-is-disrupting-air-travel-2023>).
- Dai, R., Guo, D., Han, Y., Qin, Y., 2024. Pollution-induced trips: evidence from flight and train bookings in China. *J. Dev. Econ.* 171, 103340.
- Diao, M., Zhu, Y., Zhu, J., 2017. Intra-city access to inter-city transport nodes: the implications of high-speed-rail station locations for the urban development of Chinese cities. *Urban Stud.* 54 (10), 2249–2267.
- Dobruszkes, F., Dehon, C., Givoni, M., 2014. Does European high-speed rail affect the current level of air services? An EU-wide analysis. *Transp. Res. Part A: Policy and Practice* 69, 461–475.
- Fang, H., Wang, L., Yang, Y., 2024. Competition and quality: evidence from high-speed railways and airlines. *Rev. Econ. Stat.* 1–16.
- Federal Aviation Administration (FAA). (1983). Airport capacity and delay. Accessed December, 10, 2024. www.faa.gov/regulations_policies/advisory_circulars/index.cfm/go/document.information/documentID/22824.
- Givoni, M., Banister, D., 2006. Airline and railway integration. *Transport Policy* 13 (5), 386–397.
- Givoni, M., Dobruszkes, F., 2013. A review of ex-post evidence for mode substitution and induced demand following the introduction of high-speed rail. *Transp. Res.* 33 (6), 720–742.
- Gultepe, I., 2023. A review on weather impact on aviation operations: Visibility, wind, precipitation, icing. *J. Airline Operations and Aviation Manage.* 2 (1), 1–44.
- Huang, J., Zhang, T., Sun, D., Wang, H., 2025. Urban human mobility changes based on functional areas during extreme rainstorm event: a case of Beijing “23·7” rainstorm event. *Cities* 163, 106003.
- Janic, M., 2001. Integrated transport systems in the European Union: an overview of some recent developments. *Transp. Res.* 21 (4), 469–497.
- Janic, M., 2018. Modelling the resilience of rail passenger transport networks affected by large-scale disruptive events: the case of HSR (high speed rail). *Transportation* 45, 1101–1137.
- Jiang, C., Wang, K., Wang, Q., Yang, H., 2022. The impact of high-speed rail competition on airline on-time performance. *Transp. Res. Part B: Methodological* 161, 109–127.
- Jiménez, J.L., Betancor, O., 2012. When trains go faster than planes: the strategic reaction of airlines in Spain. *Transp. Policy* 23, 34–41.
- Lavery, K.J., 1996. Economic “short-termism”: the debate, the unresolved issues, and the implications for management practice and research. *Acad. Manag. Rev.* 21 (3), 825–860.
- Lawrence, M., Bullock, R., & Liu, Z. (2019). China's high-speed rail development. World Bank Publications.
- Lee, E., Kawakita, T., Huai, Y., Lo, H.K., Zhang, A., 2024. Airline and high-speed rail collaboration and competition under travel time variability. *Transp. Res. Part A: Policy and Practice* 185, 104104.
- Li, B., Wen, C., Yang, S., Ma, M., Cheng, J., Li, W., 2024. Measuring high-speed train delay severity: static and dynamic analysis. *PLoS One* 19 (4), e0301762.
- Li, Z.C., Tu, N., Fu, X., Sheng, D., 2022. Modeling the effects of airline and high-speed rail cooperation on multi-airport systems: the implications on congestion, competition and social welfare. *Transp. Res. Part B: Methodological* 155, 448–478.
- Lin, Y., Qin, Y., Wu, J., Xu, M., 2021. Impact of high-speed rail on road traffic and greenhouse gas emissions. *Nat. Clim. Chang.* 11 (11), 952–957.
- Long, F., Zheng, L., Song, Z., 2018. High-speed rail and urban expansion: an empirical study using a time series of nighttime light satellite data in China. *J. Transp. Geogr.* 72, 106–118.
- Ma, W., Wang, Q., Yang, H., Zhang, A., Zhang, Y., 2019. Effects of Beijing-Shanghai high-speed rail on air travel: passenger types, airline groups and tacit collusion. *Res. Transp. Econ.* 74, 64–76.
- Samanta, S., Pal, D.K., Lohar, D., Pal, B., 2012. Interpolation of climate variables and temperature modeling. *Theor. Appl. Climatol.* 107, 35–45.
- Sun, X., Zhang, Y., Wandelt, S., 2017. Air transport versus high-speed rail: an overview and research agenda. *J. Adv. Transp.* 2017 (1), 8426926.
- Wan, Y., Ha, H.K., Yoshida, Y., Zhang, A., 2016. Airlines’ reaction to high-speed rail entries: Empirical study of the Northeast Asian market. *Transp. Res. Part A: Policy and Practice* 94, 532–557.
- Wandelt, S., Zhang, A., Sun, X., 2025. Global Airport Resilience Index: towards a comprehensive understanding of air transportation resilience. *Transp. Res. Part D: Transport and Environ.* 138, 104522.
- Wang, F., Wei, X., Liu, J., He, L., Gao, M., 2019. Impact of high-speed rail on population mobility and urbanisation: a case study on Yangtze River Delta urban agglomeration, China. *Transp. Res. Part A: Policy and Practice* 127, 99–114.
- Wang, K., Xia, W., Zhang, A., Zhang, Q., 2018. Effects of train speed on airline demand and price: theory and empirical evidence from a natural experiment. *Transp. Res. Part B: Methodological* 114, 99–130.
- World Meteorological Association (WMO). (2021). Weather-related disasters increase over past 50 years, causing more damage but fewer deaths. Accessed January, 5, 2025. (<https://wmo.int/news/media-centre/weather-related-disasters-increase-over-past-50-years-causing-more-damage-fewer-deaths>).
- Wu, S., Han, D., 2022. Accessibility of high-speed rail (HSR) stations and HSR–air competition: evidence from China. *Transp. Res. Part A: Policy and Practice* 166, 262–284.
- Wu, Y., Mooring, T.A., Linz, M., 2021. Policy and weather influences on mobility during the early US COVID-19 pandemic. *Proc. Natl. Acad. Sci.* 118 (22), e2018185118.
- Xia, W., Jiang, C., Wang, K., Zhang, A., 2019. Air-rail revenue sharing in a multi-airport system: effects on traffic and social welfare. *Transp. Res. Part B: Methodological* 121, 304–319.
- Xia, W., Wang, K., Zhang, A., 2018. Air transport and high-speed rail interactions in China: review on impacts of low-cost carriers, rail speed, and modal integration. *Airline Economics in Asia* 103–122.
- Xia, W., Zhang, A., 2017. Air and high-speed rail transport integration on profits and welfare: effects of air-rail connecting time. *J. Air Transp. Manag.* 65, 181–190.

- Yang, H., Zhang, A., 2012. Effects of high-speed rail and air transport competition on prices, profits and welfare. *Transp. Res. Part B: Methodological* 46 (10), 1322–1333.
- Yoo, S., Kumagai, J., Sho, K., Managi, S., 2024. Impact of high-speed railway on gentrification and regional inequality in Japan. *Transp. Res. Part D: Transport and Environ.* 134, 104346.
- Zanni, A.M., Ryley, T.J., 2015. The impact of extreme weather conditions on long distance travel behaviour. *Transp. Res. Part A: Policy and Practice* 77, 305–319.
- Zhang, A., Wan, Y., Yang, H., 2019. Impacts of high-speed rail on airlines, airports and regional economies: a survey of recent research. *Transp. Policy* 81, A1–A19.
- Zhang, Q., Yang, H., Wang, Q., 2017. Impact of high-speed rail on China's big three airlines. *Transp. Res. A Policy Pract.* 98, 77–85.
- Zheng, L., Long, F., Chang, Z., Ye, J., 2019. Ghost town or city of hope? The spatial spillover effects of high-speed railway stations in China. *Transp. Policy* 81, 230–241.
- Zheng, L., Xue, Y., Huang, D., 2025. Inter-city transport hubs and intra-city polycentric structure: evidence from high-speed rail stations and airports in China. *J. Transp. Geogr.* 123, 104132.
- Zheng, L., Wu, S., 2024. Remote high-speed rail stations, urban land supply, and the emergence of new economic activities. *Transp. Res. Part A: Policy and Practice* 189, 104226.
- Zheng, L., Zhu, H., Fan, K.Y., Chang, Z., 2024. Why high-speed rail causes heterogeneous spatial patterns in firm innovation: Perspectives from intensive and extensive margins. *PLoS One* 19 (10), e0311621.
- Zhou, L., 2024. Impact of extreme rainfall on high-speed rail (HSR) delays in major lines of China. *Weather Clim. Soc.* 16 (4), 673–680.
- Zhou, L., Chen, Z., 2020. Measuring the performance of airport resilience to severe weather events. *Transp. Res. Part D: Transport and Environ.* 83, 102362.